

Optimisation-based switch allocation to improve energy losses and service restoration in radial electrical distribution systems

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ISSN 1751-8687

Received on 11th December 2015

Revised on 7th March 2016

Accepted on 25th March 2016

doi: 10.1049/iet-gtd.2015.1509

www.ietdl.org

Abstract: This study presents a new methodology for the optimal allocation of switching devices in radial electrical distribution systems (EDSs). A specialised greedy randomised adaptive search procedure (GRASP) algorithm defines the location of a number of switching devices in order to simultaneously improve the following optimisation subproblems related to the use of the allocated switches: (i) the optimal reconfiguration of EDS and (ii) the optimal service restoration of EDS. Eventually, the objective function of the proposed switch allocation algorithm minimises the cost of the total expected energy not supplied, computed after deploying the service restoration, plus the cost of the total annual energy loss computed for every load level in a year, plus the investment costs associated with the number of installed switches. Both optimisation subproblems, i.e. the reconfiguration and the restoration of EDS, are represented by mixed-integer non-linear programming (MINLP) models and transformed into mixed-integer linear programming (MILP) models, using linearisation strategies. MILP models guarantee convergence to optimality by using convex optimisation techniques. Finally, all tests were carried out using a real 136-node distribution system, considering dispatchable and non-dispatchable distributed generation resources.

1 Introduction

The allocation of switching devices in electrical distribution systems (EDSs) is a distribution planning problem that establishes the circuits in which a number of switching devices should be installed in order to improve one or more specific objectives. The main objective of the switch allocation problem is to improve the system's reliability [1]. In practice, EDS reliability is commonly determined using customer-oriented reliability indices such as the system average interruption duration index (SAIDI), the system average interruption frequency index (SAIFI), and load-oriented indices such as the expected energy not supplied (EENS). Both customer-oriented indices, SAIDI, and SAIFI, establish the average duration and frequency of the sustained interruptions experienced by customers over a year. They are limited by EDS regulators in order to guarantee continuous electrical service. On the other hand, the load-oriented index EENS is used to quantify the restoration capacity of the system. It is suitable for measuring the performance of the restoration scheme.

Both customer-oriented reliability indices, SAIDI and SAIFI, depend on the average failure rates and restoration times of system components [2]. Since the frequency and duration of the interruptions are related to the system's topology, e.g. distant nodes are more likely to fail than the nodes nearer the substation, the allocation, and operation of switching devices is a convenient strategy to improve the EDS's reliability as well as the restoration capacity.

Though the system's reliability is the main objective of the switch allocation problem, EDS operators also use switching devices to reconfigure their networks when there are no sustained interruptions in the system, with the aim of minimising the total active power losses [3]. Thus, the decision as to whether or not to install a switching device in a circuit should not only be influenced by the restoration scheme (or the system's reliability), but also by the minimisation of the annual energy losses (AELs) calculated during continuous, non-faulty operation. The

simultaneous optimisation of the reliability indices and the EDS's operating point is a holistic approach to the classical switch allocation problem that takes advantage of the switching devices during both normal and faulty conditions.

1.1 Literature review

Owing to the combinatorial nature of the decision variables and the complex, non-analytical characteristics of the objective function, metaheuristic algorithms have been the most popular choice for solving the switch allocation problem in a reasonable amount of time and with constrained computational resources. Metaheuristics such as simulated annealing [4], genetic algorithms [5, 6], immune algorithms [7], trinary particle swarm optimisation [8, 9], ant colony optimisation [10–12], memetic algorithms [13], tabu search [14, 15], shuffled frog-leaping algorithm [16], and greedy randomised adaptive search procedure (GRASP) [17] have been successfully used to provide quality solutions for the allocation problem, even for large EDS. In addition, other specialised optimisation strategies have been proposed to solve simplified cases of the switch allocation problem, e.g. a decomposition approach [18], dynamic programming [19], and mixed-integer linear programming (MILP) [20]. Moreover, Bernardon *et al.* [21, 22] have proposed enhanced decision-making algorithms based on the conflicting and, sometimes, subjective characteristics of the problem.

Like most EDS planning and operation problems, the optimal switch allocation problem has multiple, conflicting objectives that depend on the reliability assessment methodology and the goal of the optimisation process. Customer-oriented reliability indices such as SAIDI and SAIFI have been explicitly minimised by authors in [4–6, 10, 21, 22]. Other reliability indices considered in the specialised literature are the EENS [4, 11, 13, 17, 18], the Expected Outage Cost (ECOST) [4, 7–9, 20], and other reliability costs associated with customer interruptions and unsupplied demand, calculated for each allocation proposal [7, 12, 14–16, 19].

Furthermore, most authors have also considered in the objective function the cost of purchasing, installing, and maintaining the allocated switches, i.e. the investment cost. Since the improvement of the system's reliability and the minimisation of the investment cost are conflicting objectives, most of the allocation algorithms have used multi-objective approaches to provide non-dominated solutions to the problem.

The aforementioned techniques focus on reliability enhancement, but do not take advantage of the allocated switches to diminish the AEL during normal, non-faulty operation. Furthermore, no prior work has considered the optimisation of the restoration process in order to minimise the total EENS of each allocation proposal. Similarly, the minimisation of the AEL requires the solution of a reconfiguration subproblem for each allocation proposal and load level of the system.

1.2 Contributions

This paper presents a new methodology to solve the switch allocation problem in radial EDS. The objective function simultaneously minimises the cost of the total EENS calculated after restoring the system, plus the cost of the total AEL calculated after reconfiguring the distribution system, considering the SAIFI and SAIDI limits and different load levels in a year. The total investment cost is also included in the objective function. The proposed optimisation algorithm is a modified version of the GRASP metaheuristic. To avoid a local solution, the proposed algorithm provides random initial solutions during every execution of a greedy local search. Each allocation proposal determines the set of circuits and interconnections in which switches should be installed. Both optimisation subproblems are represented by precise mixed-integer non-linear programming (MINLP) models that need to be solved for every allocation proposal. A set of linearisation strategies are used to transform the original optimisation models into MILP models. The use of MILP models guarantees convergence to optimality by using existing convex optimisation tools. Both MILP models were solved via the convex optimisation software CPLEX [23], and represented using the mathematical programming language AMPL [24]. The proposed methodology was managed by a MATLAB routine. Finally, all tests were carried out using a real 136-node distribution system, considering dispatchable and non-dispatchable distributed generation (DG) resources.

The main contributions of this paper are as follows:

- (i) A new methodology for the optimal allocation of switching devices in radial EDS with a double aim of: (a) improving the service restoration in case of any contingency and (b) minimising the AEL for different load levels, considering SAIFI and SAIDI limits.
- (ii) The use of flexible and precise MILP models for representing the service restoration and reconfiguration subproblems, considering EDS operational and reliability constraints, and dispatchable/non-dispatchable DG resources.

2 Switch allocation problem

The switch allocation problem defines the optimal position of the switching devices in the EDS, aiming to improve its operation and reliability. Let Ω_{sw} be the set that contains the branches where a switch is installed, then the proposed optimisation model for the optimal allocation of switching devices is given by (1)

$$\min_{\Omega_{sw}} f(\Omega_{sw}) = c^{EENS} \sum_{d \in \Omega_d} EENS_d + c^{AEL} \sum_{d \in \Omega_d} AEL_d + c^{switch} |\Omega_{sw}| \quad (1)$$

where:

$EENS_d$ is the EENS, calculated using (2) for every switch allocation proposal Ω_{sw} and load level d [2]. Δt_d in (2) is the number of hours at

load level $d \in \Omega_d$, 8760 is the number of hours in a year, U_{z_f} is the annual duration of the interruptions of each zone $z_f \in \Omega_z$, and $P_{z_f,d}^U$ is the amount of unsupplied demand, calculated after the service is restored following a permanent fault in the zone $z_f \in \Omega_z$. Each zone is a radial section of the EDS delimited by the allocated switches

$$EENS_d = \frac{\Delta t_d}{8760} \sum_{z_f \in \Omega_z} U_{z_f} P_{z_f,d}^U, \quad \forall d \in \Omega_d \quad (2)$$

AEL_d is the annual energy losses, calculated using (3) for every switch allocation proposal Ω_{sw} and load level d

$$AEL_d = \Delta t_d \sum_{ij \in \Omega_l} R_{ij} I_{ij,d}^2, \quad \forall d \in \Omega_d \quad (3)$$

$|\Omega_{sw}|$ is the number of elements of the set Ω_{sw} and it represents the total number of switches installed.

For each allocation proposal, the objective function in (1) requires the solution of various reconfiguration subproblems (one for each load level) in order to minimise the total AEL, and several restoration subproblems (one for each load level and zone) in order to minimise the total EENS. Therefore, the following section presents the mathematical approach used to represent and efficiently solve both subproblems related to the operation of the allocated switches.

3 Mathematical approach for the optimal reconfiguration and restoration of the EDS

EDS are designed to operate radially to simplify protection coordination and voltage regulation. Assuming a three-phase balanced system, the formulation used to calculate the steady-state operating point in this paper is based on the equations applied by the sweep load flow algorithm in [25], also called *DistFlow branch equations* [26]. The 4-node system in Fig. 1 illustrates the proposed mathematical approach. The system has two circuits, $\Omega_l = \{ij, kl\}$, one switch, $\Omega_{sw} = \{jk\}$, one dispatchable DG connected at node l , $\Omega_g = \{g\}$, and two load zones, $\Omega_z = \{1, 2\}$. As shown in Fig. 1, each zone is a radial section of the EDS, delimited by the switches; thus, every circuit and node of the system must belong to one zone defined in the set Ω_{sw} . Note that all parameters are enclosed in braces, i.e. all the demands connected to the nodes, the electrical parameters of the branches, the DG capacity limits, and the reliability rates of the zones (Λ_1 , Λ_2 , U_1 , and U_2) are parameters.

3.1 EDS operation, considering switching devices

On the basis of the *DistFlow* equations, the squared voltage drop in each circuit ij of a radial EDS is given by (4), and the square of the current magnitude in each branch is given by (5)

$$V_{i,d}^2 - V_{j,d}^2 = 2(R_{ij}P_{ij,d} + X_{ij}Q_{ij,d}) + (R_{ij}^2 + X_{ij}^2)I_{ij,d}^2, \quad \forall ij \in \Omega_l, d \in \Omega_d \quad (4)$$

$$I_{ij,d}^2 = \frac{P_{ij,d}^2 + Q_{ij,d}^2}{V_{j,d}^2}, \quad \forall ij \in \Omega_l, d \in \Omega_d \quad (5)$$

Since the impedance of the allocated switches is negligible, the voltage magnitudes in the buses of a given switch ij are equal if the switch is closed ($y_{ij,d} = 1$). Otherwise, if the switch is open ($y_{ij,d} = 0$), the difference between the nodal voltages can vary freely within their operational limits. This behaviour is mathematically represented by (6), and the square of the current

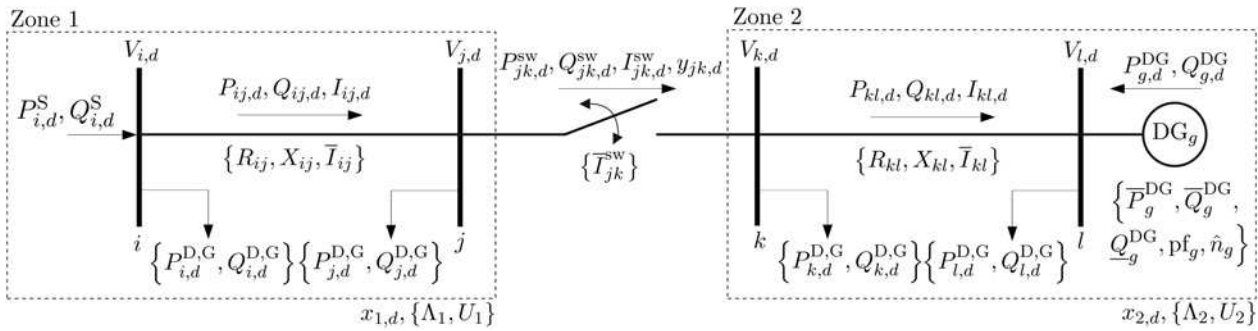


Fig. 1 Illustration of the proposed mathematical approach: a 4-node system with two zones, one switch, and a DG unit

magnitude at each switch ij is given by (7)

$$|V_{i,d}^2 - V_{j,d}^2| \leq (\bar{V}^2 - \underline{V}^2)(1 - y_{ij,d}), \quad \forall ij \in \Omega_{sw}, d \in \Omega_d \quad (6)$$

$$(I_{ij,d}^{sw})^2 = \frac{(P_{ij,d}^{sw})^2 + (Q_{ij,d}^{sw})^2}{V_{j,d}^2}, \quad \forall ij \in \Omega_{sw}, d \in \Omega_d \quad (7)$$

As shown in Fig. 1, every zone in an EDS is delimited by the switches that surround it. In the case of a permanent fault, the service restoration should isolate the faulty zone by opening the associated switches. As a result, all the electrical variables of an isolated zone are set to zero. Therefore, the binary decision variable $x_{z,d}$ is used to determine if zone z is energised ($x_{z,d} = 1$) or isolated ($x_{z,d} = 0$) after the reconfiguration/restoration. Constraints (8)–(11) define the relationship between the switch status ($y_{ij,d}$) and the zone status ($x_{z,d}$). Parameters $\hat{z}_i$ and $\hat{z}_{ij}$ determine the zone that every node i and circuit ij belong to

$$\sum_{\substack{ij \in \Omega_{sw} \\ \hat{z}_{ij} = z}} y_{ij,d} + \sum_{\substack{ji \in \Omega_{sw} \\ \hat{z}_{ji} = z}} y_{ji,d} \geq x_{z,d}, \quad \forall z \in \Omega_z, d \in \Omega_d \quad (8)$$

$$y_{ij,d} \leq x_{\hat{z}_i,d}, \quad \forall ij \in \Omega_{sw}, d \in \Omega_d \quad (9)$$

$$y_{ij,d} \leq x_{\hat{z}_{ij},d}, \quad \forall ij \in \Omega_{sw}, d \in \Omega_d \quad (10)$$

$$y_{ij,d}, x_{z,d} \in \{0, 1\}, \quad \forall ij \in \Omega_{sw}, z \in \Omega_z, d \in \Omega_d \quad (11)$$

In (8), if none of the switches that delimit zone z are closed, then zone z is isolated ($x_{z,d} = 0$). On the other hand, if even one switch is closed, then zone z is energised ($x_{z,d} = 1$) by (9) and (10). The binary nature of $y_{ij,d}$ and $x_{z,d}$ is established by (11). Since electrical power losses are computed with the term $R_{ij}I_{ij,d}^2 + jX_{ij}I_{ij,d}^2$ for each circuit $ij \in \Omega$, the expressions that guarantee AC power flow throughout the nodes, considering DG, switch operations, service restoration, and losses, are represented by (12) and (13) (see (12) and (13) at bottom of this page)

Equations (12) and (13) represent the active and reactive power balance equations for each node i and load level d . Note that the active and reactive powers at each node is given by the difference between the forecasted demand and the non-dispatchable generation of the node. The total active and reactive powers are multiplied by the zone status variable $x_{\hat{z}_i,d}$ in both (12) and (13).

$$P_{i,d}^S + \sum_{\substack{g \in \Omega_g \\ \hat{h}_g = i}} P_{g,d}^{DG} + \sum_{ji \in \Omega_l} P_{ji,d} - \sum_{ij \in \Omega_l} (P_{ij,d} + R_{ij}I_{ij,d}^2) + \sum_{ji \in \Omega_{sw}} P_{ji,d}^{sw} - \sum_{ij \in \Omega_{sw}} P_{ij,d}^{sw} = (P_{i,d}^D - P_{i,d}^G)x_{\hat{z}_i,d}, \quad \forall i \in \Omega_b, d \in \Omega_d \quad (12)$$

$$Q_{i,d}^S + \sum_{\substack{g \in \Omega_g \\ \hat{h}_g = i}} Q_{g,d}^{DG} + \sum_{ji \in \Omega_l} Q_{ji,d} - \sum_{ij \in \Omega_l} (Q_{ij,d} + X_{ij}I_{ij,d}^2) + \sum_{ji \in \Omega_{sw}} Q_{ji,d}^{sw} - \sum_{ij \in \Omega_{sw}} Q_{ij,d}^{sw} = (Q_{i,d}^D - Q_{i,d}^G)x_{\hat{z}_i,d}, \quad \forall i \in \Omega_b, d \in \Omega_d \quad (13)$$

Thus, if zone z is isolated ($x_{z,d} = 0$), there will be no power flow through it, and the zone will be de-energised.

3.2 Radiality constraint

Radial topology is guaranteed by constraint (14) and the load flow balances in (12) and (13). Equation (14) is adapted from the radiality constraint in [27] and considers the isolation of zones after the restoration process for every load level d

$$|\Omega_l| + \sum_{ij \in \Omega_{sw}} y_{ij,d} = |\Omega_b| - |\Omega_b^S| - \sum_{z \in \Omega_z} (1 - x_{z,d}) \quad (14)$$

3.3 Operational constraints

The current and voltage magnitude limits for every circuit and node of the EDS are established by constraints (15) and (16). The generation capacities of the substations are limited by constraint (17)

$$0 \leq I_{ij,d}^2 \leq \bar{I}_{ij}^2 x_{\hat{z}_{ij},d}, \quad \forall ij \in \Omega_l, d \in \Omega_d \quad (15)$$

$$\underline{V}^2 x_{\hat{z}_i,d} \leq V_{i,d}^2 \leq \bar{V}^2 x_{\hat{z}_i,d}, \quad \forall i \in \Omega_b, d \in \Omega_d \quad (16)$$

$$(P_{i,d}^S)^2 + (Q_{i,d}^S)^2 \leq \bar{S}_i^2, \quad \forall i \in \Omega_b^S, d \in \Omega_d \quad (17)$$

Note that if zone z is de-energised by the restoration process ($x_{z,d} = 0$), the current and voltages of the zone are set to zero in (15) and (16). The current limit for the switches is given by (18)

$$0 \leq (I_{ij,d}^{sw})^2 \leq (\bar{I}_{ij}^{sw})^2 y_{ij,d}, \quad \forall ij \in \Omega_{sw}, d \in \Omega_d \quad (18)$$

3.4 Dispatchable DG constraints

The active and reactive generation of each DG unit are limited by (19)–(21). As shown by (21), all DGs are limited not only by their power capacity, but also by their nominal power factor

$$0 \leq P_{g,d}^{DG} \leq \bar{P}_g^{DG}, \quad \forall g \in \Omega_g, d \in \Omega_d \quad (19)$$

$$\underline{Q}_g^{DG} \leq Q_{g,d}^{DG} \leq \bar{Q}_g^{DG}, \quad \forall g \in \Omega_g, d \in \Omega_d \quad (20)$$

$$|Q_{g,d}^{DG}| \leq P_{g,d}^{DG} \tan(\arccos(pf_g)), \quad \forall g \in \Omega_g, d \in \Omega_d \quad (21)$$

Since the proposed methodology is based on flexible mathematical optimisation models used to represent the operation of radial EDS, they can be easily adapted to consider new, different constraints such as voltage control at the dispatchable DG resources.

3.5 Reliability limits

Service restoration is an efficient procedure to minimise the EENS, but there are other customer-related reliability indices such as the SAIFI and SAIDI that utilities measure in order to guarantee suitable, continuous operation. As established in [2], the SAIFI depends on the average failure rates of the nodes ($\lambda_{k,d}$), and the SAIDI depends on the interruption duration rates of the nodes ($u_{k,d}$). Considering a system divided into load zones, each of which has unique failure (Λ_z) and duration (U_z) rates (both parameters, Λ_z and U_z , are statistical or historical data that EDS operators gather from the system), then λ_k and u_k can be calculated in terms of the system's topology, i.e. in terms of the allocated switch statuses.

3.5.1 Hypotheses: The following hypotheses are the fundamental principles of the proposed reliability constraints:

- (i) Interruptions considered in this analysis are those of long duration (more than 5 min each).
- (ii) Permanent faults in lateral fuse-protected circuits are isolated due to a fuse blow. Thus, the reliability rates of the fuse-protected nodes are not directly influenced by all the components outside the protection area of the fuse. In this case, parameters λ_{fusedk} and u_{fusedk} retain the failure and duration rates of the fuse-protected nodes. If a node is being protected by a fuse, then λ_{fusedk} and u_{fusedk} will be different from zero. Note that λ_{fusedk} and u_{fusedk} should be established based on statistical or historical information.
- (iii) Fuses located in the main feeder of each zone are disregarded from this analysis.
- (iv) Faults in one feeder do not affect the other feeders.
- (v) Switching devices, coordinated with the main breaker operation, produce the quick isolation of the zones that do not belong to the *upstream* path of the node whose reliability is being evaluated. In other words, if a fault occurs in a zone that does not belong to the *upstream* path of the node whose reliability is being

evaluated, a simple restoration scheme will restore the disconnected zone automatically.

As an example, consider the 16-node radial system shown in Figs. 2a and b. If the node whose reliability is being evaluated is node 10, the upstream path in Fig. 2a is formed by zones 4 and 1 (shaded areas). All interruptions due to faults in the rest of the zones, i.e. zones 2, 3, and 5, are quickly restored by opening a single remote-controlled switch. Otherwise, if the fault is located in any of the upstream zones, it will take nominal repair time to restore the interruption. Thus, the frequency and duration of the interruptions in node 10 in Fig. 2a are

$$\lambda_{10(a)} = \Lambda_{F1} + \Lambda_{Z1} + \Lambda_{Z4}$$

$$u_{10(a)} = U_{F1} + U_{Z1} + U_{Z4}$$

where $\Lambda_{F1, Z1, Z4}$ and $U_{F1, Z1, Z4}$ are the frequency and nominal duration of interruptions at the zones $F1$, $Z1$, and $Z4$, respectively.

Note that, as shown in Fig. 2b, if the topology of the system changes, the upstream path of node 10 also changes, and the reliability experienced by users connected to node 10 differs from that in Fig. 2a. The frequency and duration of the interruptions in node 10 in Fig. 2b are

$$\lambda_{10(b)} = \Lambda_{F2} + \Lambda_{Z2} + \Lambda_{Z4} + \Lambda_{Z5}$$

$$u_{10(b)} = U_{F2} + U_{Z2} + U_{Z4} + U_{Z5}$$

where $\Lambda_{F2, Z2, Z4, Z5}$ and $U_{F2, Z2, Z4, Z5}$ are the frequency and nominal duration of interruptions at the zones $F2$, $Z2$, $Z4$, and $Z5$, respectively.

3.5.2 SAIDI and SAIFI assessments and limits: Equations (22)–(29) determine and limit the average SAIDI and SAIFI of the EDS, according to the topology defined by the switch statuses ($y_{ij,d}$) for each load level d , and the hypotheses established above

$$\sum_{\substack{ji \in \Omega_{sw} \\ z_i = z}} f_{ji,k,d} - \sum_{\substack{ij \in \Omega_{sw} \\ z_i = z}} f_{ij,k,d} + f_{z,k,d}^S = f_{z,k,d}^D, \quad \forall z \in \Omega_z, \forall k \in \Omega_b, d \in \Omega_d \quad (22)$$

$$f_{ij,k,d} = f_{ij,k,d}^+ - f_{ij,k,d}^-, \quad \forall ij \in \Omega_{sw}, \forall k \in \Omega_b, d \in \Omega_d \quad (23)$$

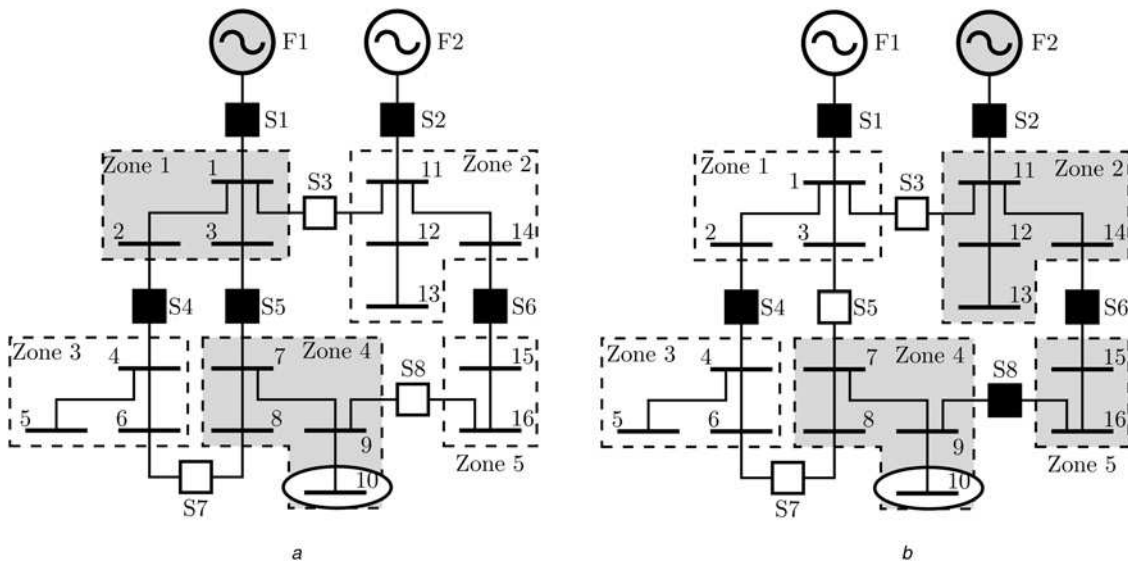


Fig. 2 Hypotheses iii. Influence of the system's topology in the reliability assessment

- a Upstream path of the node 10 is formed by zones 1 and 4
- b Upstream path of the node 10 is formed by zones 2, 4, and 5

$$f_{ij,k,d}^+ + f_{ij,k,d}^- \leq y_{ij,d}, \quad \forall ij \in \Omega_{sw}, \forall k \in \Omega_b, d \in \Omega_d \quad (24)$$

$$\lambda_{k,d} = \Lambda_{z_k} + \sum_{ij \in \Omega_{sw}} f_{ij,k,d}^+ \Lambda_{z_i} + \sum_{ij \in \Omega_{sw}} f_{ij,k,d}^- \Lambda_{z_j} + \lambda_{fusedk}, \quad \forall k \in \Omega_b, d \in \Omega_d \quad (25)$$

$$u_{k,d} = U_{z_k} + \sum_{ij \in \Omega_{sw}} f_{ij,k,d}^+ U_{z_i} + \sum_{ij \in \Omega_{sw}} f_{ij,k,d}^- U_{z_j} + u_{fusedk}, \quad \forall k \in \Omega_b, d \in \Omega_d \quad (26)$$

$$SAIFI_d = \frac{\sum_{k \in \Omega_b} \lambda_{k,d} N_k}{\sum_{k \in \Omega_b} N_k} \leq \overline{SAIFI} \quad \forall d \in \Omega_d \quad (27)$$

$$SAIDI_d = \frac{\sum_{k \in \Omega_b} u_{k,d} N_k}{\sum_{k \in \Omega_b} N_k} \leq \overline{SAIDI} \quad \forall d \in \Omega_d \quad (28)$$

$$f_{ij,k,d}^+, f_{ij,k,d}^- \in \{0, 1\}, \forall ij \in \Omega_{sw}, \quad \forall k \in \Omega_b, d \in \Omega_d \quad (29)$$

Equation (22) is the analytical formulation that discovers the *upstream* path of every zone in a radial network [28]. The identification of the *upstream* path is necessary in order to evaluate the reliability indices for each node, as established by the fifth hypotheses in Section 3.5.1. Considering an artificial demand in each node given by

$$f_{z,k,d}^D = \begin{cases} 1, & \text{if } \hat{z}_k = z \\ 0, & \text{otherwise} \end{cases}$$

the artificial flux $f_{ij,k,d}$ in (22) will differ from zero if switch ij is *upstream* of the node k , for every load level d . Using (23), the artificial flux is separated into two binary variables, where $f_{ij,k,d}^+ = 1$ if the flux is leaving node i , and $f_{ij,k,d}^- = 1$ if the flux is leaving node j . Equation (24) relates the artificial fluxes to the switch statuses $y_{ij,d}$ and guarantees that no flux will flow through an open switch. Equations (25) and (26) determine the average failure rate ($\lambda_{k,d}$) and duration of interruptions ($u_{k,d}$) of each node k and load level d , by summing the reliability parameters of the zones located in the *upstream* path of k (see hypotheses 5). Finally, $SAIFI_d$ and $SAIDI_d$ are calculated by (27) and (28), and limited by the maximum reliability indices $\overline{SAIFI}$ and $\overline{SAIDI}$ for every load level d .

Note that the former reliability constraints neither consider the effect of transferring loads among feeders, nor the direct impact of the dispatchable DG resources on the reliability assessment. Therefore, in practice, the reliability indices are expected to be even better than the topology-based estimates provided by (22)–(29).

3.6 Optimal reconfiguration of EDS

The mathematical model for the reconfiguration subproblem that minimises the AEL_d for every load level d , considering operational and reliability constraints, is represented as an MINLP problem, as shown in (30)

$$\begin{cases} \min AEL_d = \Delta t_d \sum_{ij \in \Omega_l} R_{ij} I_{ij,d}^2 \\ \text{s.t.:} \\ (4) - (29) \\ x_{z,d} = 1, \quad \forall z \in \Omega_z, d \in \Omega_d \end{cases} \quad (30)$$

The MINLP model in (30) determines the switch statuses given by the binary variable $y_{ij,d}$ and the active and reactive generation in the substations and the DGs, minimising the AEL of the system. Constraint $x_{z,d} = 1$ guarantees that all the zones will be energised by the reconfiguration process.

3.7 Optimal service restoration of EDS

The mathematical model for the optimal service restoration that minimises the $EENS_d$ in the case of a fault in zone z_f is also represented as an MINLP problem, shown in (31)

$$\begin{cases} \min P_{z_f,d}^U = \sum_{i \in \Omega_b} P_{i,d}^D (1 - x_{z_i,d}) \\ \text{s.t.:} \\ (4) - (21) \\ x_{z_f,d} = 0 \quad \forall z_f \in \Omega_z, d \in \Omega_d \end{cases} \quad (31)$$

The service restoration model in (31) determines the switch status ($y_{ij,d}$) and the generation at the substations and the DGs, in order to minimise the unsupplied active demand in the case of a permanent fault in zone z_f . Constraint $x_{z_f,d} = 0$ de-energises the zone where a permanent fault is simulated. Since the restorative state is not permanent, reliability constraints (22)–(29) are not considered.

3.8 Linearisation strategies

Both the reconfiguration problem model in (30) and the service restoration model in (31) are non-convex MINLP problems. Thus, in order to provide quality solutions for both problems in a reasonable time frame, using convex optimisation tools such as CPLEX [23], it is necessary to linearise both models. The linearisation strategies are indicated as follows.

3.8.1 Absolute value linear equivalent: Considering that the inequality $|y| \leq x$ is identical to the combined linear expressions $-x \leq y$ and $y \leq x$, the absolute values in (6) and (21) are replaced by their linear equivalents.

3.8.2 Simplification of the current magnitude: To simplify the current magnitude assessment in each circuit and switch, the squared voltage magnitudes (V_j^2) in the denominators of (4) and (7) are replaced by the squared nominal voltage of the system (V_{nom}^2), as indicated in (32) and (33)

$$I_{ij}^2 \simeq \frac{P_{ij}^2 + Q_{ij}^2}{V_{nom}^2}, \quad \forall ij \in \Omega_l \quad (32)$$

$$(I_{ij}^{sw})^2 \simeq \frac{(P_{ij}^{sw})^2 + (Q_{ij}^{sw})^2}{V_{nom}^2}, \quad \forall ij \in \Omega_{sw} \quad (33)$$

This voltage simplification has a relatively low degree of error, since the voltage magnitude at each node is limited within a narrow interval: $\underline{V}^2 \leq V_j^2 \leq \overline{V}^2, \forall j \in \Omega_b$.

3.8.3 Substitution of variables: The current and voltage magnitudes appear as squared variables in (30) and (31). Thus, the following substitutions are applied without loss of generality, where I_{ij}^{sq} , $I_{ij}^{sw,sqr}$, and V_i^{sqr} are continuous, non-negative variables

$$I_{ij}^{sq} = I_{ij}^2, \quad \forall ij \in \Omega_l$$

$$I_{ij}^{sw,sqr} = (I_{ij}^{sw})^2, \quad \forall ij \in \Omega_{sw}$$

$$V_i^{sqr} = V_i^2, \quad \forall i \in \Omega_b$$

3.8.4 Piecewise linearisation: Power flow variables P_{ij} , Q_{ij} , P_{ij}^{sw} , Q_{ij}^{sw} , P_i^s , and Q_i^s appear as both stand-alone and squared variables in (30) and (31). Therefore, it is not possible to apply the same change in variables used for the current and voltage magnitudes. To linearise the quadratic terms P_{ij}^2 and Q_{ij}^2 in (5), $(P_{ij}^{sw})^2$ and $(Q_{ij}^{sw})^2$ in (7), and $(P_i^s)^2$ and $(Q_i^s)^2$ in (17), a piecewise linear approximation is computed for each term. The piecewise linearisation is a standard procedure, similar to the

linear approximations used in [29, 30] for other power system optimisation problems.

4 Proposed switch allocation methodology

In this section, the proposed switch allocation methodology is introduced, based on the optimisation model in (1) and the mathematical models for the minimisation of the EENS and the AEL described in Section 3.

4.1 Codification

For each allocation proposal, the set Ω_{sw} contains the branches in which a switch has been installed, i.e. the allocation proposals are codified in Ω_{sw} . Switches can only be installed in existing circuit candidates ($\Omega_l^* \subseteq \Omega_l$) or interconnection paths (Ω_o) that link two different feeders. Both sets Ω_l^* and Ω_o are established by the EDS planners based on operational and constructive criteria. Switches are installed at the edges of each circuit or interconnection path. Thus, given branch $ij \in \Omega_l^*$ or Ω_o , a switch can be allocated at either edge i or j . The allocation of a switch on the edge i or j creates a new node i^* or j^* that interconnects the switch with the rest of the branch.

4.2 Neighbourhood structure

The neighbourhood structure $N_v(\Omega_{sw})$ is a function that contains all the neighbour solutions of Ω_{sw} . A neighbour solution of Ω_{sw} is an allocation proposal wherein one and only one switch of Ω_{sw} is reallocated following the branch notation. The 6-node system in Fig. 3 exemplifies the performance of the proposed neighbourhood structure. The system has four circuit candidates, $\Omega_l^* = \{A, B, E, F\}$, and two interconnection paths, $\Omega_o = \{C, D\}$. Each edge is identified by 1 or 2.

Consider that the current allocation solution is $\Omega_{sw} = \{A^1, D^2\}$, i.e. there is a switch allocated in branch circuit A (edge 1) and another switch allocated in interconnection path D (edge 2). Following the notation (not the topology) in Fig. 3, the neighbour solutions of $\Omega_{sw} = \{A^1, D^2\}$ are $N_v(\Omega_{sw}) = \{\{A^2, D^2\}; \{A^1, D^1\}; \{A^1, E^1\}\}$. The proposed neighbourhood structure is bounded by (34), which is a relatively small neighbourhood, since the number of switching devices in practice is less than the number of circuits

$$|N_v(\Omega_{sw})| \leq 2|\Omega_{sw}| \quad (34)$$

4.3 Objective function assessment

Given an allocation proposal Ω_{sw} , the objective function, $f(\Omega_{sw})$ in (1), is calculated as follows:

(i) Let $k \leftarrow 1$. On the basis of Ω_{sw} , generate the set of zones Ω_z and identify the zone that every node and circuit of the system belong to, i.e. compute $\hat{z}_i, \forall i \in \Omega_b$ and $\hat{z}_{ij}, \forall ij \in \Omega_l$.

(ii) Solve the reconfiguration problem given by (30) for every load level d , considering the linearisations in Section 3.8. If feasible, continue to step iii. Otherwise, $f(\Omega_{sw}) \leftarrow \infty$ and stop.

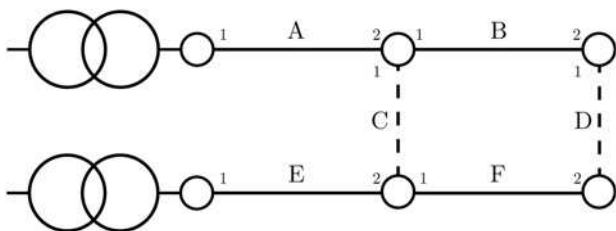


Fig. 3 6-Node test system. Neighbourhood structure example

(iii) Fix the binary decision variables in (30) and solve the resulting non-linear programming problem to correct the EDS operation point. Then $f(\Omega_{sw}) \leftarrow c^{AEL} \sum_{d \in \Omega_d} AEL_d$.

(iv) Set $z_f \leftarrow \Omega_z(k)$, where $\Omega_z(k)$ is the k th element of set Ω_z . Solve the service restoration problem given by (31) for every load level d , considering the linearisations in Section 3.8. Set

$$f(\Omega_{sw}) \leftarrow f(\Omega_{sw}) + c^{EENS} \sum_{d \in \Omega_d} \frac{\Delta t_d}{8760} U_{z_f} P_{z_f, d}^U.$$

(v) Set $k \leftarrow k + 1$. If $k > |\Omega_z|$, then set $f(\Omega_{sw}) \leftarrow f(\Omega_{sw}) + c^{switch} |\Omega_{sw}|$ and stop. Otherwise, return to step iv.

4.4 Modified GRASP algorithm

The proposed switch allocation algorithm is a metaheuristic, based on GRASP [31], which does not require a sensitivity index to construct an initial solution. Let Ω_{sw}^* be the solution produced at each iteration also known as the *incumbent solution*. Then, the pseudo-code of the proposed GRASP is shown in Fig. 4.

Lines 1 and 2 of the pseudo code initiate the incumbent solution with an empty set and an infinite value of the objective function. Loop 3–12 contain the main search procedure of the switch allocation algorithm. In this case, the stopping criterion is the maximum number of iterations of the main loop. The random function in line 4 returns an initial solution by randomly choosing elements from the sets Ω_l^* and Ω_o , considering that the number of switches in Ω_{sw} cannot be higher than $|\Omega_{sw}|$. The loop in lines 5–7 improves the initial solution Ω_{sw} by searching for neighbours with a lower objective function than Ω_{sw} . The local search in lines 5–7, also known as the *steepest descent heuristic*, stops when none of the neighbours in $N_v(\Omega_{sw})$ improve the objective function of Ω_{sw} . Finally, if the solution found by the local search is better than the current incumbent solution, it becomes the new incumbent Ω_{sw}^* . Since each iteration of the proposed switch allocation algorithm initiates with a random solution, local minima are avoided, and the diversity of the search process is guaranteed.

```

1:  $\Omega_{sw}^* \leftarrow []$ 
2:  $f(\Omega_{sw}^*) \leftarrow \infty$ 
3: while stopping criterion is not achieved do
4:    $\Omega_{sw} \leftarrow \text{Random}(\Omega_l^*, \Omega_o, |\Omega_{sw}|)$ 
5:   while a neighbor solution  $\Omega'_{sw} \in N_v(\Omega_{sw})$ 
     exists, such that  $f(\Omega'_{sw}) < f(\Omega_{sw})$  do
6:      $\Omega_{sw} \leftarrow \min\{\Omega'_{sw}\}$ 
7:   end while
8:   if  $f(\Omega_{sw}) < f(\Omega_{sw}^*)$  then
9:      $\Omega_{sw}^* \leftarrow \Omega_{sw}$ 
10:  end if
11: end while
12: return  $\Omega_{sw}^*$ 

```

Fig. 4 Pseudo-code of the proposed GRASP

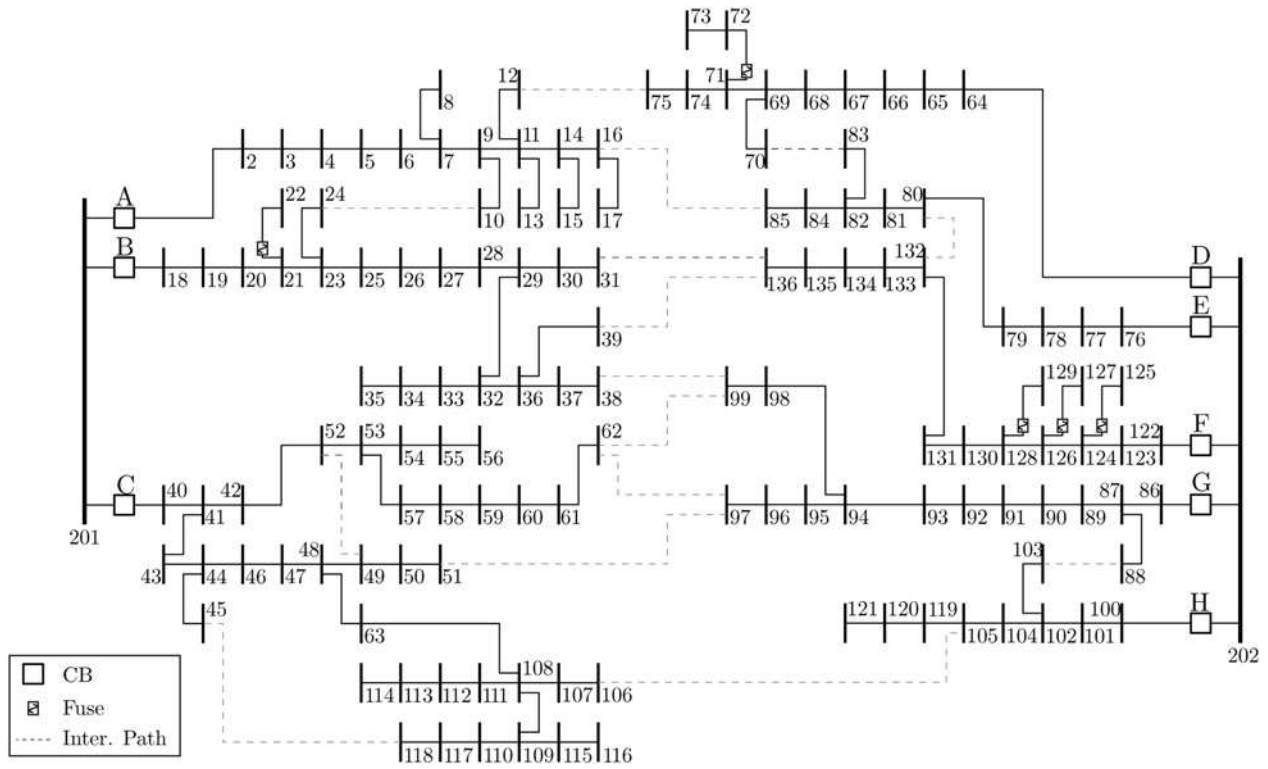


Fig. 5 Real 136-node EDS for the optimal allocation of switching devices

5 Tests and results

The 136-node EDS, shown in Fig. 5, was used to demonstrate the performance of the proposed switch allocation methodology. Electrical data can be downloaded from [32]. Non-dispatchable DGs are not considered. The constants of the system are $c^{\text{EENS}} = 15.0$ US/kWh, $c^{\text{AEL}} = 0.3$ US/kWh, $c^{\text{switch}} = 10,000$ US, $V_{\text{nomf-n}} = 7.967$ kV, $V = 0.9V_{\text{nom}}$, $\bar{V} = \frac{V_{\text{nom}}}{V}$, $\bar{I}_{ij} = 1000$ A, $\forall ij \in \Omega_{ij}$, $\text{SAIDI} = 5.24$ h/cust/year, and $\text{SAIFI} = 4.00$ f/cust/year. The system has eight radial feeders, 135 circuits, and 15 interconnection paths, indicated by dashed lines. Circuit-breakers are located at the head of each feeder and represented as squares in Fig. 5. Note that some branches are protected by upstream fuses. For all cases, the set of circuit candidates is equal to the set of circuits; thus, $\Omega_i^* = \Omega_i$. Three load levels were considered: a high level of 100% of the load during $\Delta t_1 = 1752$ h, an intermediate level of 75% of the load during $\Delta t_2 = 2628$ h, and a low level of 50% of the load during $\Delta t_3 = 4380$ h.

To simplify the λ_z and U_z assessment, an average failure ratio of $\lambda = 0.065$ failure/km/year and a resistance of $R = 0.257$ Ω /km were assumed in every circuit, and the restoration time for each zone was assumed to be $r = 2$ h/failure in each zone. Thus, for every allocation proposal, the average frequency (λ_z) and duration (U_z) of the interruptions of the zones were calculated using (35) and (36), based on the length of the circuits

$$\lambda_z = \frac{\lambda}{R} \sum_{\substack{ij \in \Omega_i \\ z_{ij} = z}} R_{ij}, \quad \forall z \in \Omega_z \quad (35)$$

$$U_z = r\lambda_z, \quad \forall z \in \Omega_z \quad (36)$$

All mathematical models were implemented in AMPL [24] and solved via CPLEX [23], with default options and a maximum gap of 1% as the optimality criterion. The modified GRASP algorithm was developed in MATLAB, using a workstation with an Intel Core i5-4570 processor. The maximum number of iterations of the algorithm was set at 300 for all tests.

5.1 Two case studies

Two case studies are described in this section. The first case (Case 1) used the switch allocation algorithm to allocate no more than ten switching devices in the 136-node EDS without considering any dispatchable DG units in the system. Thus, $|\Omega_{\text{sw}}| = 10$ and $\Omega_g = \emptyset$. Initially, the AEL of the system with neither switches nor DG units was 3.30 GWh, and the minimum voltage magnitude was 0.8989 pu at node 106 for the 100% load level, violating the V limit. The second case (Case 2) considered the operation of three dispatchable DG units as indicated in Table 1. In this case, the AEL of the system without switches was 2.99 GWh, and the minimum voltage magnitude increased to 0.9093 pu at node 85, for the 100% load level.

Table 2 shows the results obtained using the proposed methodology for both case studies. The allocation proposals in Table 2 indicate the branches and edges where switching devices needed to be installed in order to minimise the objective function in (1). The objective function of each scenario is shown in the last column of Table 2.

As expected, the value of the objective function of the allocation proposal without dispatchable DG (Case 1) is higher than the allocation proposal considering DG units (Case 2) for three reasons: (i) the operation of the DG units enhanced the voltage profile and diminished the active power losses of the system, (ii) the dispatchable DG units were used by the restoration scheme to reduce the unsupplied demand in case of a fault, and (iii) as a consequence of (i) and (ii), the number of installed switches is lower in Case 2.

Note that, for Case 1, the solution provided by the proposed methodology allocated a switch in the interconnection path 106-105. The aim was to increase the minimum voltage magnitude by keeping the switch 106-105 closed and opening another switch allocated in feeder C. Thus, it can be concluded that the proposed methodology not only improved the reliability of the system, but also optimised the steady-state operating point of the EDS.

Table 2 shows the total AEL = $\sum_{d \in \Omega_d} \text{AEL}_d$ and the total EENS = $\sum_{d \in \Omega_d} \text{EENS}_d$, calculated for each case. Both AEL and

Table 1 Parameters of the dispatchable DG units

DG [g]	Node [$\hat{n}_g$]	$\bar{P}_g^{\text{DG}}$ [kW]	$\bar{Q}_g^{\text{DG}}$ [kVar]	$\bar{Q}_g^{\text{DG}}$ [kVar]	pf_g
1	56	300	-50	200	0.9
2	116	400	-100	300	0.9
3	121	250	-30	200	0.9

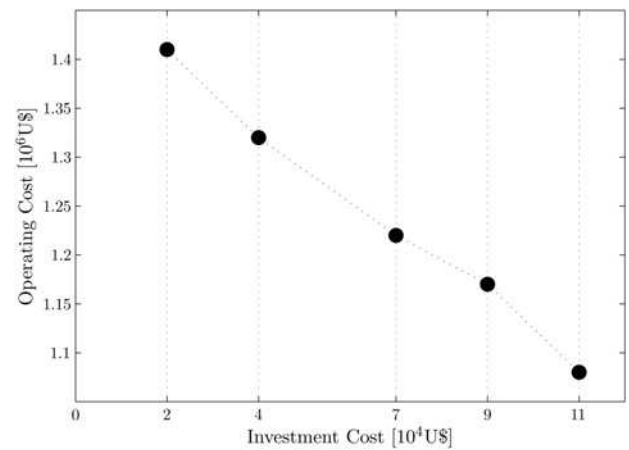
EENS are given in GWh. Notably, energy losses were much higher than the energy not supplied because the power losses are sustained over a year, whereas the EENS depends on the failure rates of each zone, which are relatively small parameters. However, in practice, the cost of the unsupplied demand is higher than the cost of the energy losses. The DG influence in Case 2 not only improved the total AEL by reducing the active power losses during normal operation, but also improved the restoration capacity as shown by the EENS value in Case 2.

With regard to the service restoration, the proposed methodology installed most of the switching devices in the feeder protected by circuit-breaker C, for all cases. As shown in Fig. 5, feeder C is the longest and most loaded feeder in the system. Also note that, as described in Table 2, the proposed methodology installed various normally closed switches in the midsection of feeders A, B, C, and D. This is common practice, since it enhances the selectivity of the service restoration scheme.

Finally, Table 3 shows the system operation during normal conditions in terms of the allocated switch statuses, the minimum voltage magnitudes, and both reliability indices, SAIDI and SAIFI, for every case and load level d , calculated using (27) and (28). Note that the topology can change according to the load level d , as long as the operational and reliability constraints are not violated.

Some conclusions can be made based on the results shown in Table 3. Since no switches were installed in feeder E and none of the DG units were connected to the nodes in feeder E, the minimum voltage magnitudes in Table 3 are identical and concentrated in node 85 (feeder E). Note that the reliability indices, SAIDI and SAIFI, for each load level are quite a bit lower than the reliability limits used throughout the simulations, ($\text{SAIDI} = 5.24$ h/cust/year and $\text{SAIFI} = 4.00$ f/cust/year). Therefore, the proposed allocation methodology focuses initially on the improvement of the voltage profile, and then on the minimisation of EENS and AEL.

An interesting scenario is the switch allocation in Case 2. As shown in Table 2, all switches were allocated in pre-existing circuits; thus, they needed to be closed in order to preserve the

**Fig. 6** Multi-objective analysis of the number of allocated switches

EDS connectivity. Other behaviours can be demonstrated, depending on the characteristics of each EDS.

5.2 Number of allocated switches

The proposed methodology includes the number of allocated switches as a term of the objective function. In this section, a multi-objective approach is used to provide non-dominated solutions, considering the conflicting relationship between the operational costs and the investment costs. The curve in Fig. 6 is built by comparing the solutions found by the proposed approach (using Case 1 from Section 5.1) and by iteratively increasing the integer parameter $|\Omega_{\text{sw}}|$, which determines the maximum number of available switches. All the non-dominated solutions found by the methodology are shown in Fig. 6. The operating costs in Fig. 6 are given by the total AEL and EENS costs, and the investment costs are given by the expression $c^{\text{switch}} |\Omega_{\text{sw}}|$.

The curve in Fig. 6 shows that a trade-off exists between the operating plus reliability costs and the investment cost. Thus, it is the responsibility of each company to establish how much money it is willing to invest in the allocation of switching devices. However, as indicated in Fig. 6, a large reduction can be obtained from installing seven switches, as a compromise solution for the utility.

Table 2 Results of the proposed switch allocation methodology: two case studies, where $\text{AEL} = \sum_{d \in \Omega_d} \text{AEL}_d$ and $\text{EENS} = \sum_{d \in \Omega_d} \text{EENS}_d$

Case 1 (without DG)				Case 2 (with DG)			
Switch locations, branches	AEL, GWh	EENS, GWh	$f(\Omega_{\text{sw}})$, 10^6 US\$	Switch locations, branches	AEL, GWh	EENS, GWh	$f(\Omega_{\text{sw}})$, 10^6 US\$
9–11 ^a , 80 ^a –81, 106–107 ^a , 64–65 ^a , 119–120 ^a , 65–66 ^a , 41–42 ^a , 105 ^a –106, 26–27 ^a	3.02	0.025	1.371	63–108 ^a , 44–45 ^a , 21 ^a –23, 101 ^a –102, 108 ^a –109, 41–42 ^a	2.99	0.020	1.257

^aEdge where the switch should be installed.

Table 3 EDS operation during normal conditions, for each load level d

Load level, %	Case 1 (without DG)				Case 2 (with DG)			
	Open switches	V_{min} , pu	SAIDI, h/cust/year	SAIFI, f/cust/year	Open switches	V_{min} , pu	SAIDI, h/cust/year	SAIFI, f/cust/year
100	106–107 ^a	0.9093	2.31	1.15	none	0.9093	1.83	0.92
75	106–107 ^a	0.9339	2.31	1.15	none	0.9339	1.83	0.92
50	105 ^a –106	0.9571	2.30	1.15	none	0.9571	1.83	0.92

^aEdge where the switch should be installed.

6 Conclusions

In this paper, a new methodology for the allocation of switching devices in radial EDS, considering operational and reliability improvement, was presented. Given a pre-determined number of switches, the proposed methodology applies a modified GRASP algorithm to simultaneously improve the total EENS and AEL of the system, considering the operation of the allocated switches and the dispatchable DG resources. For every allocation proposal, the objective function of the proposed methodology consists of the solution of various reconfiguration problems (one for each load level) and several service restorations (one for each load level and zone in the EDS), considering reliability and operational constraints.

Both the reconfiguration and the restoration subproblems were represented as MINLP models and solved using MILP solvers by applying a set of efficient linearisation strategies. A real 136-node EDS was used to allocate the switching devices considering the influence of the dispatchable DG operation in the optimisation process.

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8 Appendix

8.1 Appendix 1

The abbreviations of common terms are presented below:

AEL	annual energy losses
EDS	electrical distribution system
EENS	expected energy not supplied
DG	distributed generation
MILP	mixed-integer linear programming
MINLP	mixed-integer non-linear programming
GRASP	greedy randomised adaptive search procedure
SAIDI	system average interruption duration index
SAIFI	system average interruption frequency index

8.2 Appendix 2

The notation used throughout this paper is reproduced below for quick reference.

Sets:

Ω_b	set of nodes
Ω_d	set of load levels
Ω_g	set of dispatchable DG units
Ω_l	set of circuits
Ω_l^*	set of circuit candidates to allocate switches
Ω_o	set of interconnection paths
Ω_{sw}	set of allocated switches
Ω_z	set of load zones
Ω_b^S	set of substations

Parameters:

c^{AEL}	cost of the AEL
c^{EENS}	cost of the EENS
$f_{z,k,d}^D$	artificial flow demanded at the zone z , for every node k , and load level d . $f_{z,k,d}^D = \begin{cases} 1, & \text{if } \hat{z}_k = z, \\ 0, & \text{otherwise.} \end{cases}$
$\bar{I}_{ij}$	current limit of circuit ij
$\bar{I}_{ij}^{sw}$	current limit of switch ij

$\hat{n}_g$	node in which the dispatchable DG g is allocated
N_k	number of customers at the node k
$P_{i,d}^D$	active demand at the node i and load level d
$P_{i,d}^G$	active non-dispatchable generation at the node i and load level d
$\bar{P}_g^{\text{DG}}$	active power capacity of DG g
pf_g	power factor limit of DG g
$Q_{i,d}^D$	reactive demanded at the node i and load level d
$Q_{i,d}^G$	reactive non-dispatchable generation at the node i and load level d
$\bar{Q}_g^{\text{DG}}$	upper reactive power capacity of DG g
$\underline{Q}_g^{\text{DG}}$	lower reactive power capacity of DG g
$\bar{R}_{ij}^g$	resistance of circuit ij
$\bar{S}_i$	kVA capacity at substation $i \in \Omega_b^S$
SAIDI	maximum SAIDI of the system
SAIFI	maximum SAIFI of the system
$u_{\text{fused}k}$	duration of interruptions at the fuse-protected node k
U_z	duration of interruptions at the zone z
$\bar{V}$	upper voltage limit
$\underline{V}$	lower voltage limit
X_{ij}	reactance of circuit ij
$\hat{z}_i$	zone of the node i
$\hat{z}_{ij}$	zone of the circuit ij
z_f	faulty zone $z_f \in \Omega_z$
$\lambda_{\text{fused}k}$	frequency of interruptions at the fuse-protected node k
λ_z	frequency of interruptions at the zone z
Δt_d	duration of load level d
$ \Omega_{\text{sw}} $	maximum number of available switches

Continuous variables:

$f_{ij,k,d}$	artificial flow at switch ij , for the node k and load level d
$f_{z,k,d}^S$	artificial flow generated at the zone z , for the node k and load level d . $f_{z,k,d}^S = \begin{cases} 0, & \text{if } z \notin \Omega_b^S, \\ \text{free}, & \text{otherwise.} \end{cases}$
$I_{ij,d}$	current at circuit ij and load level d

$I_{ij,d}^{\text{sqr}}$	square of $I_{ij,d}$
$I_{ij,d}^{\text{sw}}$	current at switch ij and load level d
$I_{ij,d}^{\text{sw,sqr}}$	square of $I_{ij,d}^{\text{sw}}$
$P_{i,d}^S$	active generation at the node i and load level d
$P_{g,d}^{\text{DG}}$	active generation at DG g and load level d
$P_{z_f,d}^U$	total unsupplied demand, computed after restoring the system, in case of a permanent fault at the zone $z_f \in \Omega_z$ and load level d
$P_{ij,d}$	active flow at circuit ij and load level d
$P_{ij,d}^{\text{sw}}$	active flow at switch ij and load level d
$Q_{i,d}^S$	reactive generation at the node i and load level d
$Q_{g,d}^{\text{DG}}$	reactive generation at DG g and load level d
$Q_{ij,d}$	reactive flow at circuit ij and load level d
$Q_{ij,d}^{\text{sw}}$	reactive flow at switch ij and load level d
SAIDI_d	system average interruption duration index at load level d
SAIFI_d	system average interruption frequency index at load level d
$u_{k,d}$	interruption duration at the node k and load level d
$V_{i,d}$	voltage magnitude at the node i and load level d
$V_{i,d}^{\text{sqr}}$	square of $V_{i,d}$
$\lambda_{k,d}$	interruption frequency at the node k and level d

Binary variables:

$y_{ij,d}$	status of switch ij and load level d , where $y_{ij,d} = 1$ if switch ij is closed, or $y_{ij,d} = 0$ otherwise
$x_{z,d}$	status of the zone z and load level d , where $x_{z,d} = 1$ if the zone z is energised after the service restoration, or $x_{z,d} = 0$ otherwise
$f_{ij,k,d}^+$	positive artificial flow at switch ij , for the node k and load level d , where $f_{ij,k,d}^+ = 1$ if the flow is leaving the node i , or $f_{ij,k,d}^+ = 0$ otherwise
$f_{ij,k,d}^-$	negative artificial flow at switch ij , for the node k and load level d , where $f_{ij,k,d}^- = 1$ if the flow is leaving the node j , or $f_{ij,k,d}^- = 0$ otherwise